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CSE 4102: Computer Graphics and
Image Processing Laboratory

**VOYAGER 2 SOLAR-SYSTEM EXPLORER
AN INTERACTIVE THREE-DIMENSIONAL
COMPUTER GRAPHICS PROJECT**

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An Interactive Three-Dimensional

Computer Graphics Project

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A project report submitted for

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Abstract

Voyager 2 Solar-System Explorer is an interactive C++20 and OpenGL application that combines an educational solar-system scene with historical spacecraft playback and manual flight. Its purpose is to make computer-graphics fundamentals visible through objects that can be inspected, moved and rendered under controlled comparisons. The scene contains 26 textured celestial bodies, four planetary ring systems, a procedurally constructed Voyager 2, 8,500 instanced small bodies, a comet, orbit guides, a starfield, mission-path lines and heliosphere boundaries.

The geometry is generated from reusable indexed primitives rather than an imported spacecraft model. A shared latitude–longitude sphere provides body vertices, normals and texture coordinates. Boxes, cylinders, frustums, rods and a closed parabolic dish form 76 spacecraft assemblies containing 5,828 triangles. Scene-graph transforms preserve assembly and moon relationships; double-precision world positions and a floating origin support both spacecraft close-ups and large-scale navigation.

Offline NASA/JPL Horizons position and velocity samples drive planets and Voyager on one date through cubic Hermite interpolation. Moon revolution and axial spin use separate visual rates. The user can switch among four camera modes, inspect 18 spacecraft components, control time, select objects and pilot the probe with quaternion-based six-degree-of-freedom motion.

The lighting system implements a Sun point light, camera-attached spotlight and directional fill. Five raster shading techniques share the light model. Albedo maps, derived normal maps, an ocean specular mask and a NASA spacecraft atlas provide surface detail. Analytic sphere/ring shadow rays, a spacecraft shadow map and a hybrid Whitted view demonstrate visibility and reflection. A surface-area-heuristic triangle BVH accelerates the traced spacecraft.

Debug and Release builds, repository checks, runtime captures and input regression evidence accompany the submission. Eight uncapped Release views measured 0.64–6.60 ms per frame on the observed Radeon RX 590 at 1440 by 900. These measurements describe one machine and run. The report states the educational scale compression, visual moon phases and limited ray-tracing scope explicitly.

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CHAPTER I

Introduction

1.1 Purpose and scope

The explorer connects graphics fundamentals to inspectable results: procedural objects, hierarchical transforms, historical flybys, manual flight and raster/traced comparisons. Floating-origin precision and educational scale mappings make metre-scale hardware visible beside planets and multi-AU trajectories.

Its scene contains 26 bodies, four ring systems, 76 spacecraft assemblies with 18 inspection targets, 8,500 rocks, a comet, 7,380 stars, orbit/path guides and heliosphere boundaries. Work proceeds through geometry, hierarchy, mission/navigation, spacecraft, lighting/tracing and verification. The contribution is this integration; standard algorithms are credited, not claimed as new research.

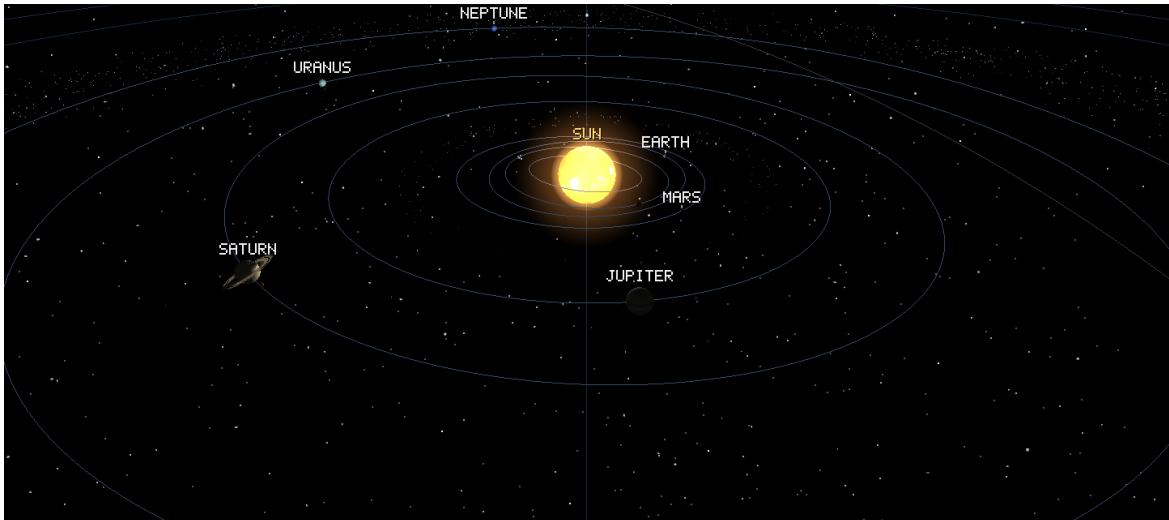


Figure 1.1: Runtime overview: labelled bodies and orbit guides organize exploration. Telemetry bands are cropped for print.

CHAPTER II

Background and System Design

2.1 From geometry to pixels

Indexed vertices hold position, normal and UV. The vertex shader transforms them; rasterization covers pixels and interpolates attributes; the fragment shader evaluates material and lighting. Depth testing preserves nearer surfaces. Transparent rings/glow blend after opaque objects without depth writes. Lines and points use separate primitive modes [1].

2.2 Architecture and data ownership

The loop is input → update → render. Window owns GLFW; Input polls events. Scene nodes own children; shared RAII meshes/textures avoid duplicate uploads. MissionController owns dated placement; camera/light controllers change viewing state. CSV catalogues separate physical data from geometry. Shaders share `lighting.glsl`.

Instancing reduces draws. Double-precision floating origin and log depth stabilize close-ups; one date keeps flybys coherent. A triangle BVH handles hardware while cheap guides remain rasterized.

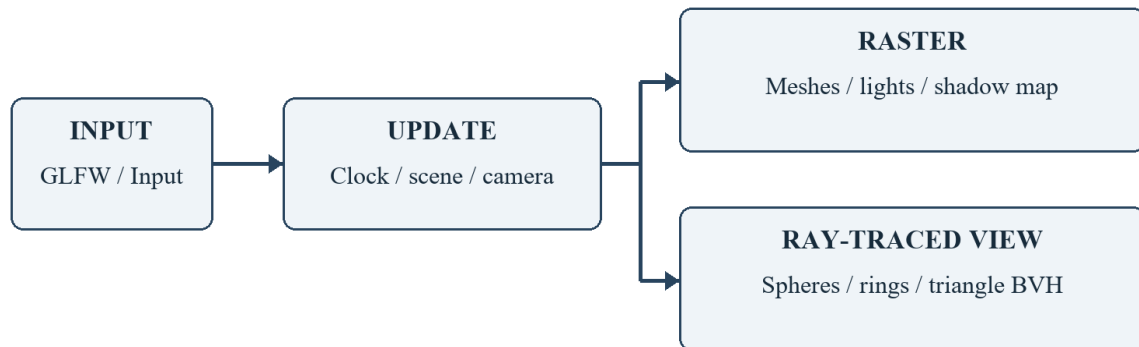


Figure 2.1: Controllers update a shared scene; raster and traced paths consume its state.

CHAPTER III

Methodology

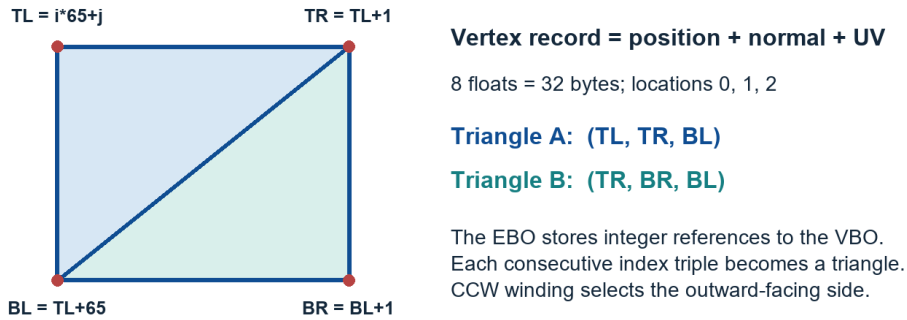
3.1 Indexed geometry and curved surfaces

Each 32-byte vertex stores position, normal and UV at locations 0, 1 and 2. A VBO holds records, an EBO unsigned indices and a VAO their interpretation. For sphere latitude i/L and longitude j/N , define $\phi = \pi i/L$ and $\theta = 2\pi j/N$:

$$\mathbf{p} = (\sin \phi \cos \theta, \cos \phi, \sin \phi \sin \theta), \quad \mathbf{n} = \mathbf{p}, \quad (u, v) = (1 - j/N, 1 - i/L). \quad (3.1)$$

Each interior cell emits (TL, TR, BL) and (TR, BR, BL) , where $TL = i(N + 1) + j$, $TR = TL + 1$, $BL = TL + N + 1$ and $BR = BL + 1$. The north/south zero-area halves are omitted. With $L = 32$, $N = 64$, $(L + 1)(N + 1) = 2,145$ vertices produce $2N(L - 1) = 3,968$ triangles and 11,904 indices. Seam positions repeat with distinct $u = 0, 1$; otherwise texture interpolation would cross the map incorrectly.

ONE INDEXED CELL / the same connectivity drives curved surfaces



Sphere poles: omit the triangle with coincident pole vertices. Seam: duplicate position, distinct $u=0 / u=1$.

Figure 3.1: One indexed sphere cell: two triangles share records and an edge. Winding is counter-clockwise from outside.

For $i = 10, j = 20$, the cell records are 670, 671, 735, 736, producing (670, 671, 735) and

(671, 736, 735). Cross products verify nonzero area and outward winding. Generator checks also verify bounds, unit normals and UV orientation before upload; a reversed winding would be culled rather than repaired by shading.

3.2 Spacecraft and environment construction

A box has 24 face-specific vertices and 12 triangles: $(b, b + 1, b + 2)$, $(b, b + 2, b + 3)$ per face. Cylinder/frustum rings use $(r_k \cos \theta, \pm h/2, r_k \sin \theta)$; side normals are proportional to $(h \cos \theta, r_b - r_t, h \sin \theta)$. Separate cap fans preserve hard normals. Both capped nonzero radii give $4n + 6$ vertices and $4n$ triangles.

Voyager is authored from boxes, a decagonal cylindrical bus, rods, frustums and a parabolic dish. Its metre dimensions share the factor 0.006 render units/metre. For dish radius R , depth D and radial coordinate r , the front profile and normal follow the surface derivative:

$$y = -D + D(r/R)^2, \quad \mathbf{n} = \text{normalize}(-2Dr \cos \theta / R^2, 1, -2Dr \sin \theta / R^2). \quad (3.2)$$

The rear surface is offset by thickness and has reversed normals/winding. Center fans, radial quads and a joined rim form a closed shell. The current 48×6 dish has 590 vertices and 1,152 triangles. Merged parts transform positions and inverse-transpose normals, then add the destination vertex offset to every appended index. A rod uses midpoint $(A+B)/2$, height $|B - A|$ and rotation from $+Y$ to $B - A$.

Each dish face has $1+6(49) = 295$ vertices and $48+5(2)(48) = 528$ triangles; the joined rim adds 96 triangles. Trusses place three rails 120° apart around the boom axis and alternating diagonal rods per bay. Open four-sided rods use eight triangles because hidden end caps contribute no visible detail.

The eighteen inspection targets reuse these families. The bus is a decagonal cylinder with blanket boxes; the high-gain antenna is a dish with curved ribs; the feed/low-gain stack uses frustums, discs and a cone; the Sun sensor is an aperture housing. The Golden Record is a disc/hub and its calibration target a plate. The magnetometer boom is a truss/canister with low-field sensor boxes at the tip. RTGs combine three cores, fins and flanges.

Plasma science uses cup frustums; cosmic ray uses a housing/telescope pair; LECP uses a

drum/platform. The scan platform carries the narrow-angle, wide-angle and IRIS barrels. Radio/plasma-wave antennas are V-shaped rods; attitude thrusters use four blocks and sixteen nozzles. Each target stores a component center and framing radius for Inspect; 18 targets group 76 assemblies rather than defining 18 individual primitives.

Table 3.1: Other primitive counts and their scene uses.

Primitive	Counts / use
Double-sided annulus	260 vertices, 256 triangles per 64-segment band
Low-poly sphere	28 vertices, 24 triangles per instanced rock
Star points / circle	7,380 points / 96 line vertices; zero triangles
Comet cone	52 vertices, 32 triangles; sphere nucleus and coma

Rings use $(r \cos \theta, 0, r \sin \theta)$, separate top/bottom normals and inner/outer edges. Uniform $\cos \phi$ avoids star polar clustering. Static rocks vary their matrices; the comet tail points anti-Sun. Orbit/path lines, six heliosphere great circles and glyph quads complete the families.

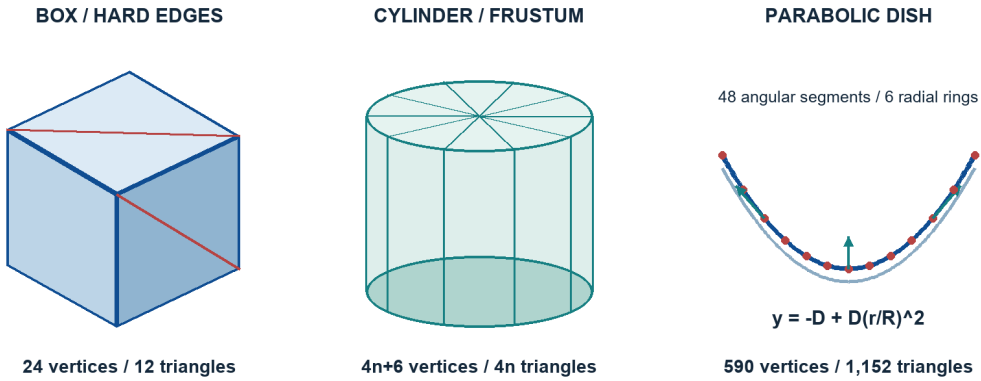


Figure 3.2: Authored construction diagrams: face-separated box, capped cylinder and the current six-ring parabolic dish.

3.3 Transforms, cameras and presentation scale

Column-vector transforms act right to left: scale in object space, rotate, then translate. Children inherit the parent matrix, so one spacecraft root moves all assemblies together:

$$M_{local} = TRS, \quad M_{world} = M_{parent}M_{local}, \quad \mathbf{n}' = \text{normalize}((M_{3 \times 3}^{-1})^T \mathbf{n}). \quad (3.3)$$

Normals use the inverse-transpose because nonuniform scale changes the perpendicular di-

rection. Planets store orbit position and axial tilt in the parent frame. Their sphere alone receives spin; moons and rings inherit the unspun frame. Moon local position and scale are divided by parent radius to compensate for inherited scaling.

For scale $(2, 1, 1)$, tangent $(1, -1, 0)$ becomes $(2, -1, 0)$. Its original perpendicular normal $(1, 1, 0)$ becomes $(0.5, 1, 0)$ under inverse-transpose, preserving a zero dot product. A local point $(1, 0, 0)$ scaled by two, rotated 90° about Y and translated $(10, 0, 0)$ becomes $(10, 0, -2)$.

$$\begin{aligned} R_{body} &= 0.50(R_{km}/6371)^{0.60}, \quad R_{Sun} \leq 6, \\ d_{render} &= 12(d_{AU}/0.387098)^{0.55}, \\ d_{moon} &= R_{parent} \left(1.6 + \sqrt{a_{km}/R_{parent,km}} \right). \end{aligned} \tag{3.4}$$

These monotonic laws preserve ordering while increasing visibility. They are display mappings, not physical scale. Before narrowing to float, the renderer subtracts the eye in double precision. Projection uses a camera-relative view and logarithmic depth $z = \log_2(1 + w)/\log_2(1 + far)$.

FreeFly moves independently; Focus orbits a body; Chase orbits in Voyager's frame; Inspect targets the craft or one of 18 components. Orbit cameras convert yaw, pitch and radius into an offset, rotate it by their frame and aim at the target. Smooth transitions avoid jumps. Picking builds a screen ray and compares angular body/spacecraft bounds.

Picking maps a cursor to $x_n = 2x/W - 1$, $y_n = 1 - 2y/H$. With field of view f , aspect a and camera basis (F, R, U) , the direction is $\text{normalize}(F + x_n a \tan(f/2)R + y_n \tan(f/2)U)$. Candidates pass an angular-radius test $\theta \leq \arcsin(r/d) + 0.021$; nearest center distance resolves selection. This forgiving test is distinct from exact traced triangle hits.

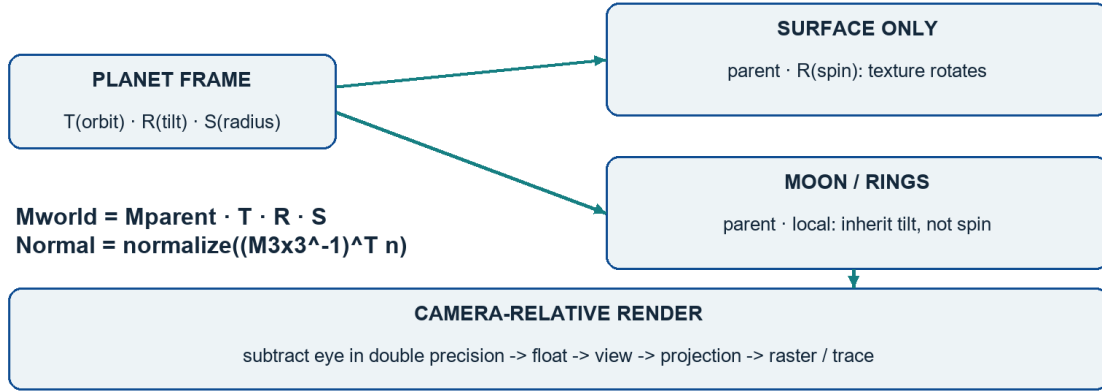


Figure 3.3: Surface spin does not rotate the planet's children. The floating origin is applied consistently to geometry, lights and ray primitives.

3.4 Historical, orbital and manual motion

Horizons rows provide Julian date, heliocentric position and velocity in AU/day [2]. With normalized interval time $s = (JD - JD_0)/h$, cubic Hermite interpolation matches both endpoints and velocities:

$$\begin{aligned} \mathbf{p}(s) = & (2s^3 - 3s^2 + 1)\mathbf{p}_0 + (s^3 - 2s^2 + s)h\mathbf{v}_0 \\ & + (-2s^3 + 3s^2)\mathbf{p}_1 + (s^3 - s^2)h\mathbf{v}_1. \end{aligned} \quad (3.5)$$

One date drives planets and Voyager. Ecliptic (x, y, z) maps to (x, z, y) before compression. Smooth flyby clearance prevents intersection with enlarged planets. After 2030, planets use $E - e \sin E = M$ two-body continuation; Voyager is ballistic. These are approximate predictions.

Hermite evaluates to p_0 at $s = 0$ and p_1 at $s = 1$; multiplying velocity by h reconciles AU/day with interval-normalized time. At $s = 0.5$, weights are 0.5, 0.125, 0.5, -0.125. Offline sampling is denser near encounters; the clock slows and substeps there to avoid skipping the flyby.

$$r = \frac{a(1 - e^2)}{1 + e \cos \nu}, \quad \nu \leftarrow \nu + \frac{2\pi(0.4)}{P_{\text{days}}} \Delta t \text{ speed}, \quad \psi \leftarrow \psi + \frac{2\pi}{P_{\text{hours}}} \Delta t \text{ speed}. \quad (3.6)$$

Moon phases and spin are visual rates; their signed periods produce retrograde cases. Moons

use constant true-anomaly speed, unlike the dated planets. The comet uses an ellipse and rotates its tail from $+Y$ toward $\text{normalize}(p_{comet} - p_{Sun})$. Manual flight uses real time, inertial translation and body-axis quaternion rotation:

$$q \leftarrow \text{normalize}(q \, q_y q_x q_z), \quad \mathbf{v} \leftarrow \mathbf{v} + \mathbf{a} \Delta t, \quad \mathbf{p} \leftarrow \mathbf{p} + \mathbf{v} \Delta t. \quad (3.7)$$

Velocity need not face forward. Braking cannot reverse it; speed is capped at 12 units/s. P pauses astronomical motion while cameras and manual flight remain usable.



Figure 3.4: Runtime Jupiter sequence: moons move independently in the tilted planetary frame; body spin does not drag their orbits.

3.5 Lighting: three types and a moving light

At surface point p , n is the unit normal and v points to the eye. A point light uses $l = (p_L - p)/|p_L - p|$; a directional light uses the same $l = -d_L$ everywhere. The Sun is a point light. The cool fill is directional and has no distance falloff. The spotlight is at the camera and follows its forward direction. Its motion therefore changes illumination when the eye moves.

$$A(d) = \frac{1}{\max(k_c + k_l d + k_q d^2, 10^{-4})}, \quad D = \max(n \cdot l, 0). \quad (3.8)$$

The optional Sun attenuation is $1/[0.3+0.7(d/20)^2]$, equal to one at $d = 20$. Default constant intensity keeps outer bodies readable. For the camera headlamp, intensity is 1.4, quadratic reach is three times nearest-surface distance, and the soft cone spans 12° – 18° :

$$C = \text{clamp} \left(\frac{(-l) \cdot d_L - \cos 18^\circ}{\cos 12^\circ - \cos 18^\circ}, 0, 1 \right), \quad I_L = \text{colour} \cdot \text{intensity} \cdot A \cdot C. \quad (3.9)$$

Only spotlights use C ; it is one for other types. With ambient 0.07, albedo a and Sun visibility

V_S , the renderer keeps the Sun and other terms separate:

$$I = a(0.07 + V_S D_S + D_{other}) + V_S S_S + S_{other}. \quad (3.10)$$

Thus an eclipse removes Sun illumination while the headlamp and fill still work. Two-sided rings use $|n \cdot l|$. K toggles lighting; $F5$, $F6$, $F7$ toggle headlamp, fill and Sun falloff. Inspect provides a camera-related studio fill to reveal hardware. Neither auxiliary light casts shadows.



Figure 3.5: Same frozen Moon and camera: Sun only, added spotlight, added directional fill. The Sun stays enabled in all three comparisons.

3.6 Shading: where and how light is evaluated

Lighting defines the source geometry; shading defines the surface normal and evaluation stage. The five $F3$ modes share diffuse $D = \max(n \cdot l, 0)$, material specular strength k_s and power p . Phong uses the reflected light direction; Blinn–Phong uses the halfway direction:

$$r = 2(n \cdot l)n - l, \quad S_P = k_s \max(v \cdot r, 0)^p, \quad h = \frac{l + v}{|l + v|}, \quad S_B = k_s \max(n \cdot h, 0)^{2p}. \quad (3.11)$$

Flat derives one face normal from $\text{normalize}(\partial_x p \times \partial_y p)$, revealing facets. Gouraud evaluates Phong lighting at vertices and interpolates terms across the triangle; highlights between vertices can be lost. Phong evaluates per pixel with an interpolated, normalized normal. Blinn–Phong uses the half-vector per pixel; doubling the exponent is this implementation’s approximate highlight-size match.

$$D_{toon} = \begin{cases} 1 & D > 0.75 \\ 0.62 & D > 0.40 \\ 0.30 & D > 0.12 \\ 0.06 & \text{otherwise} \end{cases} \quad (3.12)$$

Toon quantizes positive diffuse light into these bands, thresholds specular at 0.5 and draws rim ink when $n \cdot v < 0.22$. The unlit back side remains zero diffuse. Gouraud disables normal maps because the derived detail belongs to fragments. Unlit objects use colour/texture without illumination. Glow uses $\alpha = |n \cdot v|^p$ *opacity* and additive blending for Sun and comet shells; it is a material behavior, not a sixth selectable shading technique.

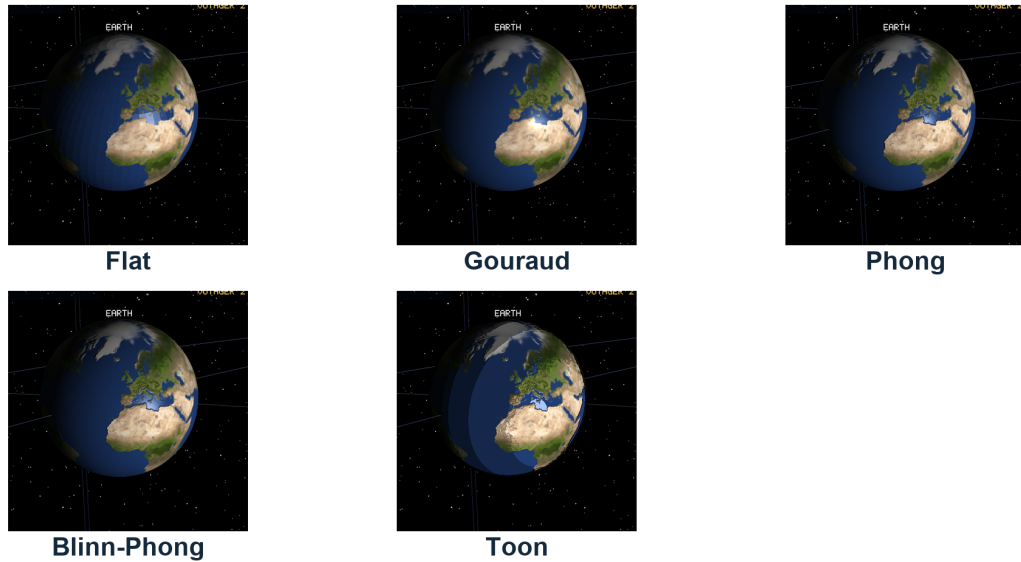


Figure 3.6: Earth in the same frozen view under Flat, Gouraud, Phong, Blinn–Phong and Toon. The presentation also shows matching spacecraft comparisons.

3.7 Textures and derived surface detail

Albedo and transforms distinguish 26 bodies. UVs duplicate the seam and preserve north/south orientation. Loading flips image rows; filtering/mipmaps reduce minification artifacts. NASA and Solar System Scope are credited [4, 5].

Voyager uses NASA hardware imagery in an atlas, not imported NASA geometry. A material selects a rectangle through $uv' = uv \cdot zw + xy$. Eleven textured finishes share the atlas; copper is untextured. Thus foil wrinkles and louvres change surface appearance while the

procedural mesh supplies the silhouette.

$$H = 0.2126R + 0.7152G + 0.0722B, \quad n_t = \text{normalize}(-s S_x(H), -s S_y(H), 1). \quad (3.13)$$

The Sobel horizontal kernel has rows $(-1, 0, 1)$, $(-2, 0, 2)$, $(-1, 0, 1)$; the vertical kernel is its transpose. Longitude wraps and latitude clamps. Encode $(n_t + 1)/2$. Screen derivatives of position/UV reconstruct the tangent frame, transforming sampled normals into the lighting frame without a tangent attribute. This colour-derived relief does not measure elevation or move vertices.

Earth's mask marks water if $B > 1.25R$, $B > 1.02G$, $R + G + B < 1.5$, assigning 255 versus 30. $F8$ toggles maps. Tracing uses $26 \ 1024 \times 512$ albedo layers and omits raster normal/specular maps.

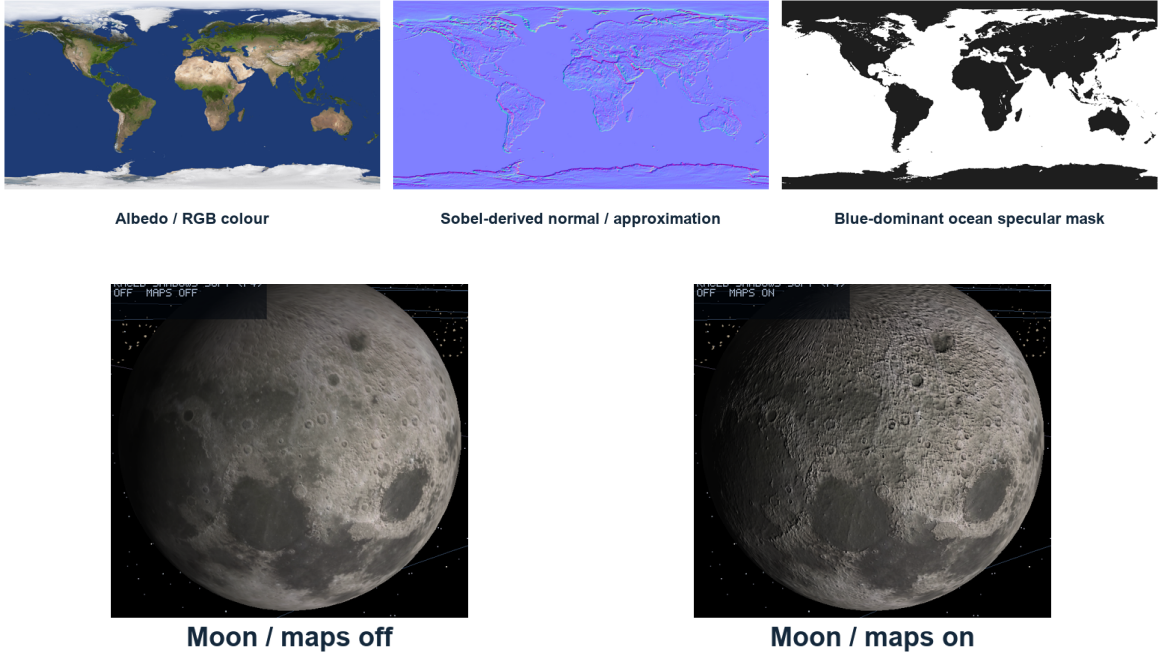


Figure 3.7: Derived Earth channels (top); runtime Moon maps off/on (bottom). Normal maps change illumination, not the silhouette.

Let p_x, p_y and u_x, u_y be screen derivatives of position and UV. The shader forms $T = (p_y \times n)u_x.x + (n \times p_x)u_y.x$ and $B = (p_y \times n)u_x.y + (n \times p_x)u_y.y$. A common inverse-length scale stabilizes T, B ; $[T \ B \ n]$ transforms the decoded map normal before normalization. Flat/Gouraud omit normal perturbation.

3.8 Shadow rays, Whitted tracing and triangle BVH

For $p(t) = o + td$ with unit d , a sphere satisfies $|p - c|^2 = R^2$. Define $b = (o - c) \cdot d$ and $f = |o - c|^2 - R^2$; its roots are $t = -b \pm \sqrt{b^2 - f}$. Reject a negative discriminant and choose the nearest positive root. A ring intersects its plane:

$$t = \frac{(c - o) \cdot N}{d \cdot N}; \quad R_{in} \leq |o + td - c| \leq R_{out}. \quad (3.14)$$

Raster Sun rays test spheres and annuli. Hard mode tests the Sun center; soft uses finite-Sun disc overlap for analytic penumbrae. Rings multiply visibility by $1 - \text{opacity}$. Voyager's raster self-shadows use a 2048^2 Sun depth map and 3×3 PCF depth comparisons, avoiding per-pixel BVH traversal.

$F9$ finds the nearest primitive. For triangle edges $e_1 = B - A$, $e_2 = C - A$, Möller–Trumbore defines $P = d \times e_2$, $\det = e_1 \cdot P$, $Q = (o - A) \times e_1$ [6]:

$$u = \frac{(o - A) \cdot P}{\det}, \quad v = \frac{d \cdot Q}{\det}, \quad t = \frac{e_2 \cdot Q}{\det}. \quad (3.15)$$

Accept $t > 0$, $u, v \geq 0$, $u + v \leq 1$. Weights $(1 - u - v, u, v)$ interpolate UV/normal. Voyager's 12-bin SAH BVH has 5,828 triangles, 6,723 nodes and depth 24. Local-space rays traverse a 32-entry stack near-first; bounding-box slab intervals reject empty regions.

For each box axis, compute the ray parameters of its two slab planes and sort them. The box entry is the maximum lower bound; exit is the minimum upper bound. Reject disjoint intervals or entry beyond the closest hit. SAH evaluates splits using child surface area times triangle count. Near-first traversal tightens the closest distance; leaves test at most four triangles.

Whitted lighting uses Blinn–Phong, Sun rays, four ring layers and one optional reflection $d_r = d - 2(d \cdot n)n$ [7], weighted by reflectivity. Reflections omit Voyager. The environment stays rasterized with shared log depth. Bias prevents self-hits; refraction and diffuse global illumination are absent.

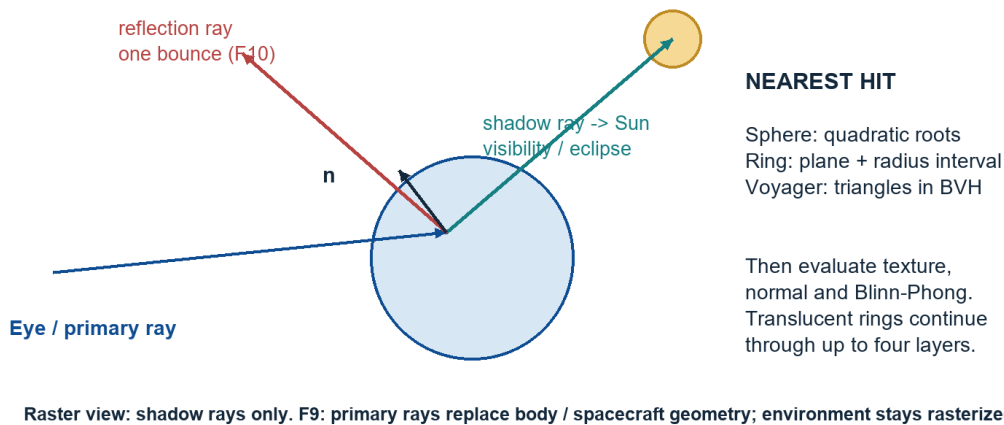


Figure 3.8: Primary, shadow and reflection rays have different jobs. Tracing is hybrid, with analytic bodies and a triangle-BVH spacecraft.

3.9 Instancing, visibility and render-pass control

Each rock field uploads one mesh plus instance matrices at attributes 3–6 with divisor one: matrices advance per instance, vertices per vertex. Three draws cover 8,500 rocks, but $8,500 \times 24 = 204,000$ triangles still require processing. Sphere LOD selects 224 triangles for $r/d < 0.012$, 960 below 0.05 and 3,968 otherwise. These are angular proxies, not fixed pixel thresholds.

Normalized frustum planes reject a bounding sphere when $n \cdot (c - eye) + w < -r$. Children are still visited because a moon can remain visible outside its parent's bounds. Rendering submits opaque surfaces first; transparent rings use source-alpha blending and glow uses additive blending, both with depth tests but no depth writes. HUD glyphs are two-triangle screen-space quads. Shared materials/resources and cached uniforms reduce repeated CPU setup; the next chapter measures the resulting workload.

The static spacecraft BVH is built once, not reconstructed on every frame. OpenGL 3.3 texture buffers carry nodes and triangle attributes to GLSL; camera-relative rays transform into craft-local space before traversal. Keeping the transformed direction unnormalized preserves the ray parameter under that linear transform. The traced pass writes compatible logarithmic depth, permitting raster guides and environmental objects to share visibility. Separate albedo layers supply body colour, while the raster material path retains derived normal/specular maps. This division preserves inspectable hardware without tracing thousands of explanatory rocks and lines.

CHAPTER IV

Implementation, Results and Discussion

4.1 Verification and performance

Debug/Release builds, both repository checks, input regression and captures passed. Sticky polling preserves brief taps [3]. Captures compare lights, maps, shading, shadows and tracing. Manual driver/input coverage is incomplete; existing linker warnings remain.

Table 4.1: Release timings: RX 590, 1440 by 900, vsync off; milliseconds.

View	Frame	GPU	View	Frame	GPU
Overview	0.64	0.42	Voyager inspect	0.99	0.58
Launch chase	0.83	0.65	Dish close-up	1.15	0.96
Jupiter focus	0.68	0.51	Saturn traced	2.10	1.77
Saturn focus	0.72	0.55	Voyager traced	6.60	5.89

The eight-view mean is 1.71 ms for one machine/run. GPU queries can wait; frame/GPU timings differ. Counters omit shadow-depth, tracing and HUD draws, although GPU timing includes HUD. Section 3.9 explains the distinct optimization costs.

CHAPTER V

Conclusions

The explorer integrates procedural geometry, shared transforms, dated playback, lighting, shading, textures and hybrid tracing. Compressed scales, visual motion, static fields, approximate hardware and derived relief limit fidelity. Tracing omits Voyager reflections, raster lighting maps, refraction and global illumination; post-2030 motion is predictive. Measured relief, dated moons, articulation and wider shadow support require future profiling.

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